

3D Printing – Terms & Definitions

A working glossary for learning and quick reference. Add anything you want explained.

Tessellated (STL Tessellation)

Made of many small flat shapes (usually triangles) that approximate a curved surface.

DETAIL

When you export a 3D model as an **STL**, the computer doesn't store true curves. Instead, every curved surface is broken into many tiny flat triangles.

Those triangles form a *tessellation* — a tiling of shapes that fit together with no gaps. The smaller and more numerous the triangles, the smoother the surface appears.

This is why zooming in closely on a curved STL shows visible facets.

Full-size graphics for tessellation examples are available in the library.

These pictures show exactly what “tessellated” means in 3D modeling:

- A smooth shape
- Broken into many flat triangles
- All touching edge-to-edge with no gaps
- Forming a “mesh” that approximates the curve

That's what every STL is made of under the hood.

Parametric (Parametric Modeling)

A modeling approach where shapes are controlled by dimensions and rules instead of fixed geometry.

DETAIL

In a parametric model, sizes such as width, height, wall thickness, hole diameter, or spacing are defined as parameters. Changing a parameter automatically updates the entire model.

This allows one design to generate many variations without redesigning from scratch. Once a model is exported as an STL, it is no longer parametric, because the parameters are baked into fixed triangles.

Vase Mode (Spiralize Outer Contour)

A printing mode where the object is printed as one continuous spiral wall with no infill, no top layers, and no seams.

DETAIL

In Vase Mode, the slicer generates a single continuous extrusion path that slowly rises in the Z direction instead of printing discrete layers. This eliminates layer changes, Z-seams, infill, and top surfaces.

Because only one wall is printed: - Filament usage is greatly reduced - Print time is often shorter - The object is hollow and lightweight

Vase Mode is commonly used for decorative or non-structural objects such as vases, lamp shades, planters, and stylized trees.

Limitations: - Very weak structural strength - No internal features or solid areas - Not suitable for functional or load-bearing parts

Different slicers may use different names: - Bambu Studio / PrusaSlicer: *Spiralize Outer Contour* - Cura: *Vase Mode*